

Controlling surface reactions of hydrogen atoms by the electron spin

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Gas molecules possess various internal degrees of freedom. Whereas recent advances in surface dynamics have clarified the roles of translational, vibrational and rotational motion in gas-surface reactions, the effect of its electron spin on surface reactions remains poorly understood. Here, we demonstrate the spin dependence of H adsorption and abstraction reactions on a Ni surface using a spin-polarized atomic-hydrogen beam. The results indicate that H adsorption is suppressed when the H electron spin is oriented perpendicular to the magnetization direction of the Ni(111) surface. Furthermore, the same spin preference is observed for the H abstraction reaction on a D-covered surface. These findings suggest that controlling the spin orientation of molecules and surfaces by applying an external magnetic field provides a viable means to modulate surface reaction rates.

An intriguing feature of a particle is its spin-degree of freedom. The symmetry with respect to the permutation of identical particles imposes rigorous restrictions on the wavefunction and energy level of the relevant system. Whereas nuclear spins lead to the presence of nuclear-spin isomers that are strongly coupled with the rotational motion¹, electron spins induce exchange interaction that plays an important role in chemical processes including surface reactions. The exchange interaction due to electron-nucleus Coulomb attraction favors the spin-singlet state, which corresponds to chemical-bond formation as described by the Heitler-London theory². In contrast, the exchange interaction originating from electron-electron repulsion tends to stabilize a spin-triplet state in accordance with Hund's rule². These two effects would compete in gas-surface interactions. Compared with other degrees of freedom related to nuclear motion, such as vibrational and rotational motion, the role of electron spin in the gas-surface interaction has been less explored to date.

The effects of the electron spin on surface reactions have attracted considerable attention in recent years. Theoretical calculations on ammonia synthesis on magnetic surfaces revealed that the spin-split d-band plays an important role in the reactions^{3–5}. With respect to

reactant gas molecules, the influence of the spin state of oxygen molecules and atoms on the adsorption were examined in detail experimentally. The ground states of both O₂ (³Σ_g[−]) and O(³P). It has been reported that the adsorption probability of O₂ molecules depends on the relative orientation of the O₂ spin and substrate magnetization of Ni and other magnetic surfaces^{6,7}. Moreover, the adsorption probability of O(¹D) is substantially higher than that of O(³P) on a graphite surface⁸.

A hydrogen atom consists of an electron and a proton, each having spin 1/2. As the simplest molecule, the adsorption of hydrogen on surfaces is regarded as a prototype reaction and is of fundamental importance in many fields, ranging from heterogeneous catalysis, interstellar chemistry to plasma catalysis, which has been extensively studied on various surfaces to date^{9–11}. In the dissociative adsorption of H₂, the breaking of the H–H bond is a key process, which is promoted by vibrational excitation as well as translational motion¹². Furthermore, because the potential energy surface depends on the molecular orientation, the influence of the rotational motion on adsorption has been reported for several metal surfaces^{13–18}. For atomic H adsorption, energy dissipation has been a central topic over the past decade.

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Recent advanced experimental and theoretical studies using a well-defined atomic H beam have clarified that electron-hole pair excitations play a decisive role in the H adsorption on metal surfaces^{19,20}. On the other hand, the behavior of the H spin was theoretically analyzed during scattering at a Cu surface²¹. Although spin effects in H adsorption can be anticipated, experimental evidence about the spin effect has yet to be established.

In this study, we investigate spin effects in H adsorption on the ferromagnetic Ni(111) surface. By controlling the spin direction of H atoms relative to the Ni surface, whose magnetization lies in the surface plane, using a recently developed spin-polarized H beam²², we compare the adsorption behavior for cases in which the H spin is either parallel or perpendicular to the Ni majority spin. As a result, we find that H adsorption is substantially suppressed when the H spin is oriented perpendicular to the Ni surface. Furthermore, the H abstraction reaction, which is well known as a prototypical non-thermal reaction proceeding via the Rideal-Eley / hot-atom mechanism^{23–25}, is also suppressed when the H spin is oriented perpendicular to the Ni surface. The magnitude of the spin effect depends on the strength of the applied magnetic field which controls the electron spin direction with respect to the surface. The present results strongly suggest controlling the spin direction provides a means to modulate the rates of surface reactions.

Results and discussion

Figure 1a shows a schematic of the experimental approach to investigate the spin effect on the adsorption and abstraction of hydrogen on surfaces. By applying a magnetic field in the direction perpendicular (parallel) to the surface with a permanent magnet around the sample, the spin direction of the incident H atom was aligned perpendicular (parallel) to the surface. Note that the effect of the magnetic field due to the Ni sample on the H spin direction is discussed later. To avoid the blocking of the beam by the magnet, the Ni crystal was rotated by 45° relative to the H beam direction (Fig. 1b). After H beam irradiation at a surface temperature (T_s) of 89 K, H₂ thermal desorption spectroscopy (TDS) was carried out to quantify the amount of the surface H coverage. Figure 2a shows a series of H₂ TD traces measured after H beam irradiations for 60, 360 and 900 s, respectively. Each figure contains 4 traces: Two traces are obtained after H beam irradiation under an external magnetic field (H) of 79 mT applied either perpendicular or parallel to the surface. Also shown in the figures are the spectra measured without applying a magnetic field. The H₂ TD spectrum exhibits two peaks (β_1 and β_2) depending on the H coverage. The β_2 peak at

380 K saturates at a coverage of 0.5 monolayer (ML) forming a 2×2 structure with H atoms adsorbed at threefold hollow sites. Above 0.5 ML, the desorption temperature is lowered as recognized by the β_1 peak due to repulsive interaction between adsorbed H atoms^{26,27}.

As shown in Fig. 2a, the amount of adsorbed H is clearly larger when the external magnetic field is applied in the surface-parallel direction (parallel- H configuration) compared with that of the perpendicular direction (perpendicular- H configuration). It is worth emphasizing that no such difference between the two configurations was observed on the nonmagnetic Cu(111) surface (see Fig. 2b). Furthermore, the amount of adsorbed H in the parallel- H configuration is comparable to that measured without an external magnet installed around the Ni sample. The latter corresponds to the H spin alignment by a stray field of 0.12 mT originating from the magnet located 90 mm below the Ni sample, with the magnetic field direction parallel to the surface (no-MG configuration).

It should be noted that H adsorption through dissociative chemisorption of residual H₂ in the vacuum chamber cannot be neglected^{28,29}. To evaluate the effect of the residual H₂ adsorption, TDS was performed after the surface was left without H beam irradiation for the same durations as those in the irradiation experiments. After subtracting the contribution of the residual H₂ adsorption, the ratios of the H coverage after H beam irradiation in the parallel- H configuration to those in the perpendicular- H and no-MG configurations are plotted in Fig. 2c. Whereas the coverage ratio between the parallel- H and perpendicular- H configurations exceeds 7 at the zero-coverage limit, it approaches -1 at long H irradiation times. The former value unambiguously represents the spin effect in the H adsorption at the bare Ni surface. The latter value can be understood by considering the limited number of available sites near the saturation coverage of 1 ML. Although the adsorption probability differs between the two configurations, the resultant H coverage eventually reaches 1 ML for both configurations. It should be noted that when the surface is covered by H atoms the abstraction reaction between the incident and adsorbed H atoms occurs in addition to H adsorption. The measured surface H coverage therefore reflects the net results of these two competing reaction channels. In contrast to the coverage ratio between the parallel- H and perpendicular- H configurations, the ratio between the parallel- H and no-MG configurations remains nearly constant over the entire H beam irradiation time.

Turning to the H abstraction reaction, Fig. 3a shows the time evolution of the HD yield when the D-covered Ni(111) surface was exposed to the H beam. The Ni(111) surface was initially covered with

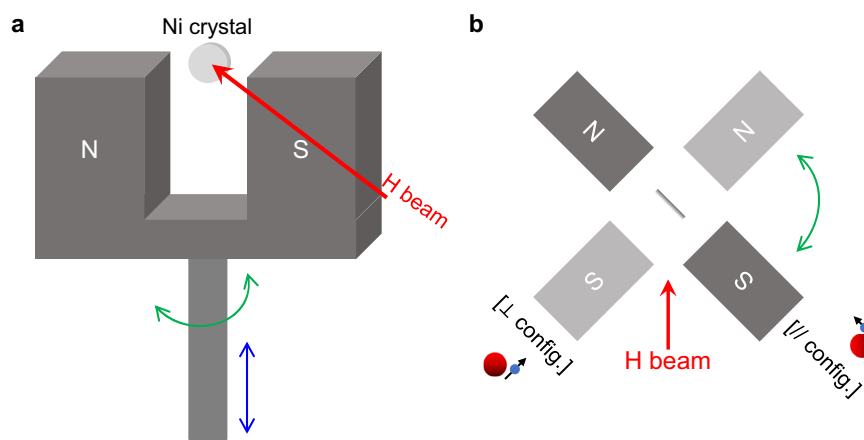


Fig. 1 | Schematic of the experimental conditions. **a** Side view of the magnet and sample arrangement. The red arrow represents the H beam. The magnet can be moved in the directions indicated by the blue and green arrows, allowing control of the magnitude and direction of the magnetic field around the Ni crystal. **b** Top view

of the magnet and sample arrangement. The magnet is positioned either parallel or perpendicular to the sample surface, referred to as the parallel ($\parallel$) and the perpendicular ($\perp$) configurations, respectively.

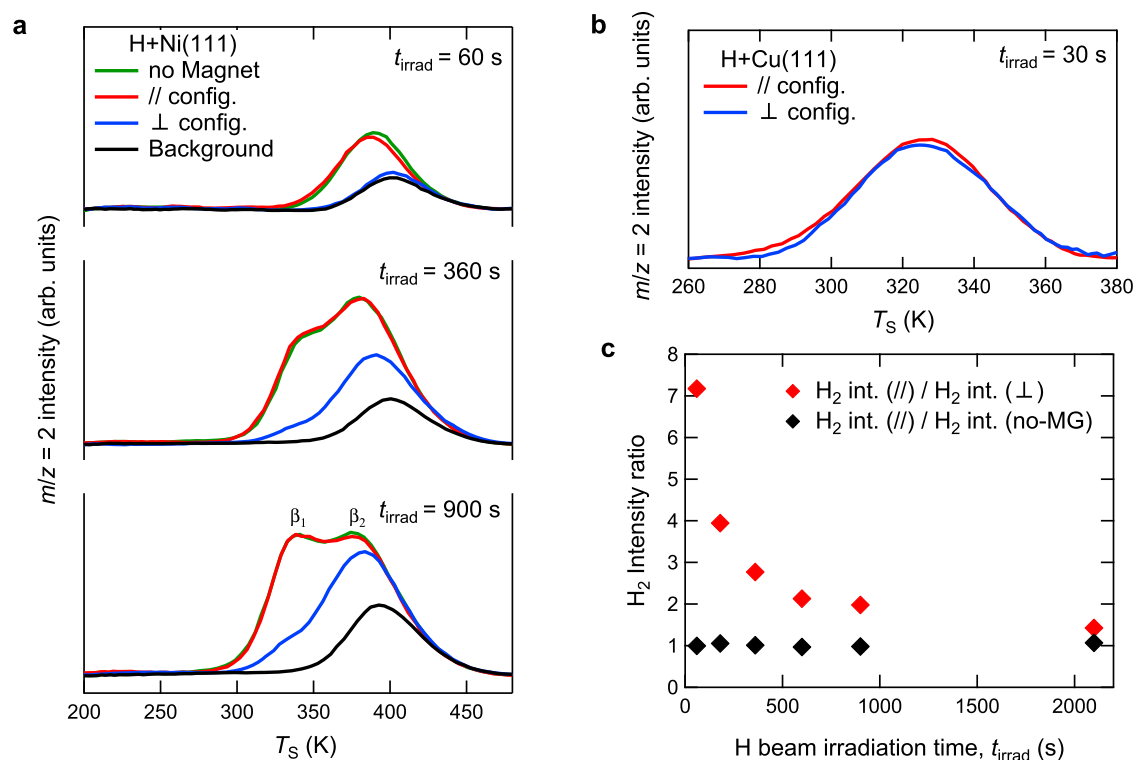


Fig. 2 | Hydrogen atom adsorption on Ni(111) and Cu(111). **a** A series of H_2 TD traces measured after H beam irradiation on Ni(111) at $T_s = 89$ K for 60, 360 and 900 s. Each figure contains 4 traces measured after H beam irradiation under an external magnetic field (H) of 79 mT perpendicular ($\perp$, blue) and parallel (//, red) to the surface, without installing a magnet around the sample (no Magnet, no-MG, green), and after the surface was left without H beam irradiation for the same

duration as the irradiation experiment (Background, black). **b** H_2 TD traces measured after H beam irradiation on Cu(111) at $T_s = 200$ K for 30 s under an external magnetic field of 180 mT perpendicular (blue) and parallel (red) to the surface. **c** Ratios of the H coverage after H beam irradiation in the parallel- H configuration to that in the perpendicular- H configuration (red diamonds), as well as to that in the no-MG configuration (black diamonds).

1 ML of D atoms prepared by backfilling the vacuum chamber with D_2 (purity 99.6%) at a pressure of 1.3×10^{-4} Pa for 20 min. Under these conditions, H beam irradiation induces HD formation at the surface³⁰. The direction of the external magnetic field was alternately changed between orientations parallel and perpendicular to the surface by adjusting the magnet direction around the sample. The HD yield in the parallel- H configuration is clearly higher than that in the perpendicular- H configuration while the magnitude of the difference depends on the magnitude of the external magnetic field. In contrast, the HD yield decreases monotonically with H beam irradiation regardless of the external magnetic field direction. This gradual decrease in the HD yield with increasing H beam irradiation is attributed to the consumption of surface D atoms.

Figure 3b shows the initial ratio of the HD yield at $t = 0$ between the parallel- H and perpendicular- H configurations as well as the yield ratio between the parallel- H and no-MG configurations. The HD yield at $t = 0$ for each configuration was obtained by extrapolating the linear relation obtained from the fit to the first 4 data sets in Fig. 3a. It is found that the ratio of the HD yield between the parallel- H and perpendicular- H configurations increases drastically with increasing strength of the magnetic field. It should be mentioned that HD formation is observed even if the Ni(111) is out of the H beam as seen in the end part of the time evolutions of the HD yield for the top and bottom traces in the figure. This is most likely due to the reaction of H with D adsorbed on the chamber wall (BG reaction). The abstraction reaction coefficient is reported to be approximately 0.12 in a previous study³⁰, and H atoms scattered from the D-covered Ni(111) surface potentially react with D on the chamber wall. This background contribution was taken into account for the determination of the ratio of the HD yield between the two configurations shown in Fig. 3b.

We discuss the origin of the dependence of the H reactivity on the magnetic-field direction with respect to the surface. We first address the magnetic properties of the Ni surface. The magnetism of the bare Ni(111) surface has been investigated by magnetic imaging techniques in detail³¹. It was reported that the magnetization at the surface is entirely in-plane in the absence of an external magnetic field. Based on the study by Dzemiantsova et al.³¹, the magnetization at the surface is expected to remain in-plane regardless of the direction of the applied magnetic field, either parallel or perpendicular to the surface, although domain walls may move depending on the magnitude and direction of the magnetic field applied in the present study. In addition, the magnetization of Ni gives rise to a magnetic field extending into the vacuum region above the surface. Since the saturation magnetization of Ni is about 600 mT³² and the magnetic permeability of Ni is 100–600^{33,34}, the magnetic flux density outside the Ni surface of each magnetic domain is estimated to be 1–6 mT with its direction in-plane because of the boundary condition that the parallel component of the magnetic field must be continuous across the surface.

We next consider the situation in which an H atom approaches the Ni surface. Under the no-MG configuration, there exist magnetic domains and the H spin tends to be aligned in the direction parallel to the Ni magnetization direction of each domain. However, it should be noted that there is a possibility that the H spin becomes antiparallel to the Ni spin when H atoms are incident to magnetic domains with reversed magnetization. Because the magnetic field direction is expected to be reversed within a length scale comparable to the magnetic-domain size (1–3 μm) above the surface³¹, the field rotation speed ($\sim 10^9$ /s) for an H atom with a velocity of 3000 m/s is comparable to or faster than the Larmor precession frequency of the H spin at the magnetic field. This implies that the H spin possibly becomes

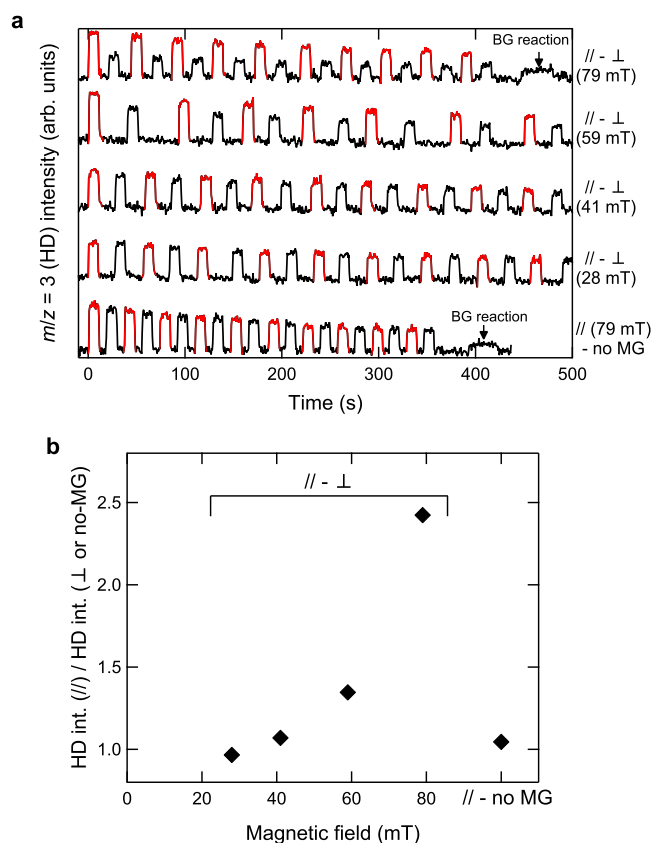


Fig. 3 | H abstraction reaction on D-covered Ni(111). **a** Time evolution of the HD yield where the magnetic field was switched alternately between orientations parallel and perpendicular to the surface (or no-MG configuration). Red and black colors correspond to H beam irradiation under the parallel and perpendicular (or no-MG) configurations, respectively. The different timing of the field direction change is due to the manual adjustment of the magnet position. **b** Initial HD yield ratios at $t = 0$ between the parallel and perpendicular configurations plotted as a function of the applied magnetic field strength, as well as the yield ratio between the parallel ($B = 79$ mT) and no-MG configurations (black diamonds).

antiparallel without following the local magnetic field adiabatically. When a high external magnetic field is applied parallel to the surface, the surface magnetization forms a single domain aligned with the external field, and the H spin is also aligned in the direction of the external magnetic field. In contrast, when the external magnetic field is applied perpendicular to the surface, the Ni magnetization remains in-plane and the H spin is aligned in the direction of the sum of the external field and the magnetic field generated by Ni.

The present experimental results shown in Fig. 2c indicate the adsorption probability of H with its spin oriented parallel to the Ni spin is higher than that with the perpendicular orientation. This observation is counterintuitive, because a spin-parallel configuration is generally unfavorable for chemical-bond formation between H and Ni atoms. As discussed in Supplementary Note 1, the preference for the parallel spin configuration suggests that exchange interactions arising from the electron-electron Coulomb repulsion between the electron in the H atom and those in the surface Ni atoms play an important role in the observed spin dependence. One possible interpretation of the spin-dependent sticking is that an additional repulsive interaction may give rise to an energy barrier in the potential-energy curve (PEC) for the perpendicular spin configuration, thereby effectively reducing the adsorption probability.

We further consider the possible magnitude of the energy barrier for the perpendicular spin configuration. Since the mean translational energy of the H beam in the present study is as low as 66 meV (the

normal component of the incidence energy is 33 meV at an incidence angle of 45° , Supplementary Fig. 1)²², if a spin-dependent barrier exists, its height is expected to be on the order of several tens of meV, roughly in the range of 10–100 meV. Previous studies have reported very high adsorption probabilities for non-spin-polarized H atoms on Ni(111). A sticking probability of approximately 0.9 was observed for H atoms with an incidence energy of 0.99 eV produced by photolysis¹⁹, and an initial sticking probability close to unity was reported for H atoms produced by a heated W tube (2000 K)^{30,35}. Although the incidence energy was not reported in the latter study, it is expected to be substantially higher than that of the present H beam (66 meV). These previous reports are therefore not inconsistent with the present interpretation, because H atoms with such high kinetic energies would readily overcome an energy barrier of the magnitude proposed here.

As in the case of H adsorption, the abstraction reaction was also suppressed when an external magnetic field is applied perpendicular to the surface, although the effect was weaker as shown in Fig. 3b. This suggests that the surface magnetization remains in-plane for a D-covered Ni(111) surface, and that there is an energy barrier for incident H atoms with a perpendicular spin orientation. According to a previous theoretical study³⁶ and our density-functional theory (DFT) calculations (see Supplementary Fig. 2 and 4), the magnetic moment per surface Ni atom is reduced by about 16–30% upon H adsorption. We therefore speculate that the barrier height for the D-covered surface is lower than that for a clean surface as a result of the reduced magnetic moment. Such an influence of adsorbed H may also be relevant to H adsorption on Ni(111) as evidenced by the decrease in the intensity ratio between the parallel-H and perpendicular-H configurations with increasing H coverage as shown in Fig. 2c. Moreover, as the magnetic-field strength decreases, the ratio of the HD yield decreases as shown in Fig. 3b. This is most likely because the barrier height is reduced in the perpendicular-H configuration; the spin moment of H becomes increasingly inclined toward the in-plane direction as the resultant magnetic field given by the sum of the external field and the field due to Ni tilts toward the in-plane direction.

To gain insight into the mechanism of the energy barrier, we performed preliminary DFT calculations for the spin-dependent PECs. Although these calculations do not show an explicit adsorption barrier, they indicate that the PECs depend on the relative spin orientation between H and Ni (see Supplementary Fig. 2 and 4). Further theoretical investigations are required to establish an atomic-level understanding of the observed spin-dependent adsorption. In particular, it will be necessary to determine spin-dependent potential-energy curves while preserving the spin orientation and moment of the H atom. Such calculations will clarify whether a spin-dependent adsorption barrier exists on the potential-energy curves.

Even if a spin-dependent barrier is found on these PECs, this would not necessarily explain the experimental observations. During the approach of the H atom to the surface, the H spin orientation might change, which corresponds to transitions between spin-dependent PECs. The role of transition between spin-dependent PECs has been discussed theoretically in the context of O_2 adsorption on Si³⁷ and Al³⁸ surfaces; the incident O_2 molecule is a spin-triplet state whereas the final adsorption state is a low-spin state. In those studies, non-adiabatic transitions between spin states were proposed to influence the adsorption process, due to the absence of a barrier in the calculated spin-adiabatic PECs available at that time. We note, however, that subsequent studies employing screened hybrid approaches^{39–41} suggested that such non-adiabatic spin-state transitions may not be essential for describing O_2 adsorption on Al surfaces. Likewise, whether transitions between spin-dependent PECs play an important role in the present H/Ni(111) system remains an open question and warrants further theoretical investigation.

Although an unambiguous interpretation of the present results has not yet been established, the present study demonstrates that the

spin degree of freedom plays a significant role in chemical reactions at surfaces and that controlling the spin direction of molecules and surfaces provides a promising means to modulate surface reaction rates. To clarify the general role of spin in surface reactions, a precise evaluation of the surface magnetization including domain size and magnetization direction under an external magnetic field is required. In addition, further investigations on other magnetic surfaces such as Co and Fe would be highly valuable. Furthermore, measuring the incidence-energy dependence of adsorption would allow us to quantitatively evaluate the barrier height for H atoms. These issues will be addressed in future studies. We also hope that the present study stimulates further theoretical work to elucidate the origin of the observed spin-dependent effects in surface reactions.

Methods

Experimental methods

The experiments were carried out in an ultrahigh vacuum chamber at a base pressure of 3.0×10^{-8} Pa equipped with a spin-polarized H beam source²². The spin-polarized H beam was prepared with a combination of a microwave discharge source consisting of a quartz tube with a diameter of 10 mm and a subsequent beam collimation by a skimmer with a diameter of 1 mm and a hexapole magnet. As a feed for the hydrogen discharge, high purity H₂ (7N) was used. To assist and enhance the discharge, water of about 1% was mixed with the H₂ gas. After reporting ref.²², a center stop (2 mm in diameter) was installed at the entrance of the hexapole magnet to block non-dissociated H₂ and H₂O and the down-spin component of H at the central region of the hexapole magnet. In this setup, the main component of the H beam is the up-spin state of an H atom. With trajectory simulations, the spin polarization at the sample position taking account of the velocity distribution and velocity-dependent focusing is estimated to be 51% as $(I_u - I_d)/(I_u + I_d)$, where I_u and I_d denote the intensities of the up- and down-spin states. The diameter of the up-spin H beam spot at the sample surface is ≤ 8 mm at normal incidence.

The clean Ni(111) surface was obtained by Ar⁺ sputtering with an energy of 1 keV at room temperature, annealing at 1100 K, followed by oxidation in 1.3×10^{-5} Pa of O₂ and reduction in 1.3×10^{-4} Pa of H₂. The Cu(111) surface was cleaned by repeated cycles of Ar⁺ sputtering followed by annealing to 750 K. The surface cleanliness and crystallographic order were verified by TD spectra of H₂^{27,42–45} and low energy electron diffraction, respectively. A permanent magnet was positioned nearly parallel to the $[110]$ direction for the parallel-*H* configuration, whereas it was positioned perpendicular to the surface, corresponding to the $[111]$ direction, for the perpendicular-*H* configuration. The magnetic-field strength at the sample position was controlled by adjusting the magnet position (the height relative to the sample), which was measured prior to installing the sample. The magnetic field strength at the sample position was set at 79 mT for the adsorption experiment on Ni(111), while it was varied in the range of 28–79 mT for the abstraction experiment. While under the external magnetic field applied the electron spin polarization of the H beam remains to be higher than 37%, under the no-MG condition the polarization is reduced to 26–28%⁴⁶. Although the magnetization is in-plane at the Ni(111) surface, aligning the H spin antiparallel to the Ni spin was difficult in the current setup. This is because, when applying an external magnetic field parallel to the surface, the spin direction of Ni and H atom (here the up-spin state is solely dealt for the sake of simplicity) are both aligned in the same direction as the external field. The experiments for H adsorption on Cu(111) were carried out with the magnetic field of about 180 mT. The amount of adsorbed H was quantified by post-TDS following H beam irradiation. On the other hand, the amount of the reaction product HD was monitored by a quadrupole mass spectrometer (QMS) during the H beam irradiation for the abstraction experiment. As shown in Fig. 1, in the abstraction experiment, the direction of the external magnetic field was changed

alternately either parallel or perpendicular to the surface. It should be noted the QMS sensitivity was found to slightly change due to the magnetic field from the magnet depending on the magnet position. Therefore, the sensitivity change was corrected by measuring the dependence of the background H₂ signal ($m/z = 2$) on the magnet direction. The H adsorption and abstraction experiments on Ni(111) were carried out at $T_s = 89$ K, whereas the H adsorption experiment on Cu(111) was conducted at $T_s = 200$ K. It should be mentioned that, because the experiments on Cu were performed before installing the center stop, the surface temperature was maintained above the water desorption temperature in order to prevent the adsorption of H₂O originating from the H beam source.

Theoretical calculation

DFT calculations were carried out with the VASP-code version 6.3.2^{47–51}. We adopted the projector-augmented wave formalism^{52,53}, with the Perdew-Burke-Ernzerhoff generalized gradient approximation exchange correlation functional⁵⁴ and the electronic convergence of 10^{-7} eV and 450 eV for the cut off energy of the plane wave basis set. We adopted the Monkhorst-Pack method⁵⁵ to perform the Brillouin zone integrations with $5 \times 5 \times 1$ k-points. The 3 d and 4 s electrons of Ni and the 1 s electron of H were treated as valence electrons, while the remaining electrons were accounted for as core electrons by using pseudo-potentials. We modeled the clean Ni(111) surface as a periodic slab with a size of $3 \times 3 \times 3$ separated by a vacuum region of 20.0 Å. We confirmed that the results are essentially unchanged for 4-layer and 5-layer slabs. We corrected the interaction between adjacent Ni surfaces in the periodic boundary condition by putting the dipole moment at the middle of the supercell with the dipole correction method⁵⁶. We considered the lattice constant 3.524 Å for fcc-Ni, and the atomic structure of Ni(111) surfaces was optimized by minimizing the atomic force within 0.001 eV/Å. For PEC, the total energy relative to the energy at a sufficiently distant place (5 Å) from Ni(111) is calculated as a function of the distance of H from the surface at fcc-hollow, hcp-hollow, bridge and on-top sites by including the spin-orbit interaction. In the calculations, the magnetization direction of the Ni(111) surface was fixed to the $[2\bar{1}\bar{1}]$ direction (in-plane direction of the surface). Then, we considered three directions of H spins, which are the $[\bar{1}\bar{1}\bar{1}]$ direction (perpendicular to the surface) and two in-plane directions on the surface, the $[2\bar{1}\bar{1}]$ direction (parallel to the Ni spin) and the $[\bar{2}11]$ direction (antiparallel to the Ni spin). For the calculation of the spin-dependent potential, the initial orientation of the H spin was set to be perpendicular, parallel or antiparallel to the Ni majority spin, and the electronic structure of the system was calculated in a self-consistent manner. In the calculations, the spin directions and moments of all atoms were fully relaxed. During the calculations, the orientation of the H spin and the spin moments of H and Ni remained unchanged in some cases, whereas in other cases they spontaneously changed. For the calculation of local spin moments, the sphere sizes of H and Ni are set to 0.37 Å and 1.286 Å, respectively.

Data availability

The data supporting the findings of this study are available within the article and the Supplementary Information. Source data are provided with this paper as a Source Data file. Source data are provided with this paper.

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Author contributions

K.F. conceived and supervised the project. H.U. performed the experiments on the H adsorption and abstraction measurements on Ni surfaces. H.N. performed the experiment of the H adsorption on Cu surfaces. Y.O., Z.-D.S., and T.O. assisted in part of the H adsorption experiments on Ni surfaces. F.W. performed the H-trajectory simulations. H.U. and H.N. analyzed the experimental data. Y.M. and W.A.D. performed the DFT calculations. H.U. Y.M., and K.F. wrote the manuscript and supplementary materials in discussion with T.O.

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Competing interests

The authors declare no competing interests.

Additional information

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